

Remifentanil: Drug information - UpToDate

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For additional information see "Remifentanil: Patient drug information" and "Remifentanil: Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

Special Alerts

Opioid Labeling Safety Updates August 2025

The FDA is requiring safety labeling changes for all opioid pain medications to better highlight the risks of long-term use, including addiction, misuse, and overdose. This decision follows a public advisory meeting that reviewed data from 2 large postmarketing studies, which revealed serious risks and a lack of evidence supporting long-term opioid effectiveness. The updated labels will include clearer risk information, stronger dosing warnings, revised guidance on appropriate use, details on overdose reversal agents, and drug interactions. They will also stress safe discontinuation to prevent harm, outline new risks like overdose-related brain injury and digestive issues, and strengthen warnings about interactions with CNS depressants, including gabapentinoids. Additionally, the FDA is requiring a new clinical trial to further assess long-term opioid use.

Further information may be found at: <https://www.fda.gov/drugs/drug-safety-and-availability/fda-requiring-opioid-pain-medicine-manufacturers-update-prescribing-information-regarding-long-term>

ALERT: US Boxed Warning

Addiction, abuse, and misuse:

Because the use of remifentanil exposes patients and other users to the risks of opioid addiction, abuse, and misuse, which can lead to overdose and death, assess each patient's risk prior to prescribing and reassess all patients regularly for the development of these behaviors and conditions.

Life-threatening respiratory depression:

Serious, life-threatening, or fatal respiratory depression may occur with use of remifentanil, especially during initiation or following a dosage increase. To reduce the risk of respiratory depression, proper dosing and titration of remifentanil are essential.

Risks from concomitant use with benzodiazepines or other CNS depressants:

Concomitant use of opioids with benzodiazepines or other CNS depressants, including alcohol, may result in profound sedation, respiratory depression, coma, and death. Reserve concomitant prescribing of remifentanil and benzodiazepines or other CNS depressants for use in patients for whom

alternative treatment options are inadequate.

Brand Names: US

- Ultiva

Pharmacologic Category

- Analgesic, Opioid;
- Anilidopiperidine Opioid

Dosing: Adult

General anesthesia

General anesthesia:

Induction: Continuous infusion: **IV:** 0.5 to 1 mcg/kg/minute; if endotracheal intubation is to occur in <8 minutes, an initial dose of 1 mcg/kg may be administered over 30 to 60 seconds.

Coronary bypass surgery: Continuous infusion: **IV:** 1 mcg/kg/minute.

Maintenance:

Intermittent: **IV:** 1 mcg/kg administered every 2 to 5 minutes.

Continuous infusion: **IV:** Consider increasing concomitant anesthetics with infusion rate >1 mcg/kg/minute. Infusion rate can be titrated upward in increments of 25% to 100% or downward in decrements of 25% to 50% every 2 to 5 minutes.

With nitrous oxide (66%): **IV:** 0.4 mcg/kg/minute (range: 0.1 to 2 mcg/kg/minute).

With isoflurane: **IV:** 0.25 mcg/kg/minute (range: 0.05 to 2 mcg/kg/minute).

With propofol: **IV:** 0.25 mcg/kg/minute (range: 0.05 to 2 mcg/kg/minute).

Coronary bypass surgery: Continuous infusion: **IV:** 1 mcg/kg/minute (range: 0.125 to 4 mcg/kg/minute); supplemental dose: 0.5 to 1 mcg/kg.

Postoperative recovery/Postanesthesia care unit

Postoperative recovery/Postanesthesia care unit (ie, immediate postoperative period):

Note: Bolus doses are not recommended.

Continuous infusion: **IV:** 0.1 mcg/kg/minute (range: 0.025 to 0.2 mcg/kg/minute). Infusion rate may be adjusted every 5 minutes in increments of 0.025 mcg/kg/minute. Infusion rates >0.2 mcg/kg/minute are associated with respiratory depression.

Coronary bypass surgery, continuation as an analgesic into the ICU: Continuous infusion: **IV:** 1 mcg/kg/minute (range: 0.05 to 1 mcg/kg/minute).

Analgesia during monitored anesthesia care

Analgesia during monitored anesthesia care:

Note: Supplemental oxygen is recommended.

Single IV dose administered 90 seconds prior to local anesthetic:

Remifentanyl alone: **IV:** 1 mcg/kg over 30 to 60 seconds.

With midazolam: **IV:** 0.5 mcg/kg over 30 to 60 seconds.

Continuous infusion beginning 5 minutes prior to local anesthetic:

Remifentanyl alone: **IV:** 0.1 mcg/kg minute.

With midazolam: **IV:** 0.05 mcg/kg/minute.

Continuous infusion administered after local anesthetic:

Remifentanyl alone: **IV:** 0.05 mcg/kg/minute (range: 0.025 to 0.2 mcg/kg/minute).

With midazolam: **IV:** 0.025 mcg/kg/minute (range: 0.025 to 0.2 mcg/kg/minute).

Note: Following local or anesthetic block, infusion rate should be decreased to 0.05 mcg/kg/minute; rate adjustments of 0.025 mcg/kg/minute may be done at 5-minute intervals. Infusion rates >0.2 mcg/kg/minute are associated with respiratory depression.

Mechanically ventilated patients in the ICU, pain and sedation

Mechanically ventilated patients in the ICU, pain and sedation (off-label use):

Loading dose: **IV:** 1.5 mcg/kg once.

Maintenance dose: Continuous infusion: **IV:** 0.008 to 0.25 mcg/kg/minute (**or** 0.5 to 15 mcg/kg/hour) (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling; however, remifentanyl pharmacokinetics are unchanged in patients with end stage renal disease.

Dosing: Liver Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling; however, remifentanyl pharmacokinetics are unchanged in patients with severe hepatic impairment.

Dosing: Obesity: Adult

The recommendations for dosing in patients with obesity are based upon the best available evidence and clinical expertise. Senior Editorial Team: Jeffrey F. Barletta, PharmD, FCCM; Manjunath P. Pai, PharmD, FCP; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC.

Class 1, 2, or 3 obesity (BMI ≥ 30 kg/m²):

Use **adjusted body weight** for initial weight-based dose calculations (based on indication), then titrate to clinical effect (Ref). Clinicians should **not** change dosing weight from one weight metric to another during therapy (ie, adjusted body weight to/from actual body weight) (Ref). Refer to "Dosing: Adult" for indication-specific doses.

Rationale for recommendations:

Studies evaluating other opioids in patients with obesity have shown large variations in opioid requirements with no relationship to actual body weight (Ref). Pharmacokinetic studies evaluating remifentanyl, a highly lipophilic compound, have shown pharmacokinetic parameters are more closely related to lean body weight (LBW) or fat-free mass than to actual body weight. These studies suggest dosing based on ideal body weight or corrected LBW for target-controlled infusion systems (Ref).

Guidelines have suggested using LBW for initial weight-based dosing of opioids, titrated to effect (Ref); however, there are concerns with calculation errors when using the complex LBW formula, especially in critical situations (Ref). There are also concerns with underdosing in clinical situations where rapid sedation is needed (eg, induction of anesthesia, ICU sedation) (Ref). Use of an alternative size descriptor (ie, adjusted body weight) is recommended for dosing in patients with class 1, 2, or 3 obesity (BMI ≥ 30 kg/m²) (Ref).

Dosing: Older Adult

Decrease initial dose by 50% and cautiously titrate to effect. Refer to adult dosing.

Dosing: Pediatric

(For additional information see "Remifentanyl: Pediatric drug information")

Note: For use of continuous IV infusion in obese patients, dose should be based on ideal body weight (IBW) in obese patients (>30% over IBW).

Analgesia, postoperative; mechanically ventilated patient

Analgesia, postoperative; mechanically ventilated patient: Limited data available: Children and Adolescents 3-16 years: Continuous IV infusion: 0.1 mcg/kg/minute with or without adjunctive medication, titrate to effect; dosing based on a randomized, double-blind comparative trial of remifentanyl versus fentanyl in 22 postoperative orthopedic spinal surgery patients requiring mechanical ventilation (remifentanyl group, n=11; age: Mean:13 years, range: 3-16 years); similar efficacy and adverse effect profile were reported with both treatment groups (Ref).

Analgesia/Sedation in mechanically ventilated patients

Analgesia/Sedation in mechanically ventilated patients: Limited data available: Infants ≤ 2 months: Continuous IV infusion: Initial: 0.15 mcg/kg/minute, titrate to effect; dosing based on a randomized, double-blind, comparative study of neonates and young infants (n=23; GA: ≥ 36 weeks; PNA: < 60 days) requiring mechanical ventilation who were given a combination of either midazolam/remifentanyl (n=11) or midazolam/fentanyl (n=12); mean remifentanyl effective dose: 0.23 mcg/kg/minute; maximum reported dose: 0.5 mcg/kg/minute; efficacy data and adverse effect profile were similar to fentanyl(Ref).

Anesthesia, maintenance of anesthesia

Anesthesia, maintenance of anesthesia: Continuous IV infusion:

Infants 1-2 months: Maintenance of anesthesia with nitrous oxide (70%): 0.4 mcg/kg/minute (range: 0.4-1 mcg/kg/minute); supplemental bolus dose of 1 mcg/kg may be administered, smaller bolus dose may be required with potent inhalation agents, potent neuraxial anesthesia, significant comorbidities, significant fluid shifts, or without atropine pretreatment

Infants ≥ 3 months, Children, and Adolescents: Limited data available: Maintenance of anesthesia with halothane, sevoflurane, or isoflurane: 0.25 mcg/kg/minute (range: 0.05-1.3 mcg/kg/minute); supplemental bolus dose of 1 mcg/kg may be administered every 2-5 minutes. Consider increasing concomitant anesthetics with infusion rate > 1 mcg/kg/minute. Infusion rate can be titrated upward in increments up to 50% or titrated downward in decrements of 25% to 50%. May titrate every 2-5 minutes.

Anesthesia, total intravenous; with or without propofol induction

Anesthesia, total intravenous (TIVA); with or without propofol induction: Limited data available: Infants, Children, and Adolescents: IV: Loading dose (may omit if propofol induction used): 0.5 mcg/kg/minute for 3 minutes (total dose: 1.5 mcg/kg) or 1 mcg/kg over 1 minute; followed by an initial maintenance dose: 0.05-0.1 mcg/kg/minute; titrate in 0.05 mcg/kg/minute increments every 3 minutes to effect; usual effective range: 0.2-0.5 mcg/kg/minute; has been used in combination with propofol (bolus and/or infusion) (Ref).

Endotracheal intubation for elective procedures

Endotracheal intubation for elective procedures: Limited data available:

IV: Infants, Children, and Adolescents: Usual range: 1-3 mcg/kg/dose, doses as high as 4 mcg/kg have been used. Dosing based on multiple trials in patients undergoing elective procedures requiring intubation who were not receiving neuromuscular blockers and remifentanyl administered in combination with other induction agents such as propofol, ketamine, or sevoflurane (Ref).

Intranasal (using parenteral 100 mcg/mL formulation): Children ≤ 7 years: 4 mcg/kg/dose; administer half of the total dose in each nostril; wait 2-3 minutes before attempting intubation. Dosing based on a double-blind, randomized, controlled trial (n=188), remifentanyl was administered after sevoflurane induction and overall, was found more effective than saline placebo at both endpoints of 2 minutes and 3 minutes postdose. Results also showed that within the remifentanyl treatment groups, good or excellent intubation conditions were reported more often at 3 minutes postdose than at 2 minutes (91.7% vs 68.2%) (Ref).

Procedural sedation

Procedural sedation: Limited data available, dosage not established: Infants, Children, and Adolescents: IV: 0.5 mcg/kg with propofol has been used to induce analgesia and adjunct sedation prior to esophagogastroduodenoscopy (n=22; age range: 2-12 years) and short hemato-oncologic invasive procedures (n=30, age range: 2-18 years) (Ref). A higher dose of 3 mcg/kg was used in already sedated (sevoflurane) infants and young children who required apnea for CT or MRI imaging (n=12, age range: 6-16 months) (Ref). In a dose-finding trial, continuous IV infusion of remifentanyl at 0.2 mcg/kg/minute has been shown more effective than a lower 0.1 mcg/kg/minute in 60 children 2-12 years undergoing diagnostic cardiac catheterization; pretreatment with oral midazolam and local groin anesthetic also utilized (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

Removed by hemodialysis (30%). There are no dosage adjustments provided in the manufacturer's labeling; however, remifentanyl pharmacokinetics are unchanged in patients with end stage renal disease.

Dosing: Liver Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling; however, remifentanyl pharmacokinetics are unchanged in patients with severe hepatic impairment.

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified. Adverse reactions reported in adults unless otherwise indicated.

>10%:

Cardiovascular: Hypotension (4% to 29%)

Dermatologic: Pruritus ($\leq$ 18%)

Gastrointestinal: Nausea (neonates, infants, children: 8%; adults: $\leq$ 44%), vomiting (neonates, infants, children: 12%; adults: $\leq$ 22%)

Nervous system: Headache ($\leq$ 18%)

Neuromuscular & skeletal: Muscle rigidity (1% to 17%; chest wall rigidity: 5%)

1% to 10%:

Cardiovascular: Bradycardia (1% to 7%), flushing (1%), hypertension (1% to 2%)

Dermatologic: Diaphoresis (6%)

Local: Localized warm feeling (1%), pain at injection site (1%)

Nervous system: Agitation ($\leq$ 1%), chills (1%), dizziness ($\leq$ 5%), shivering (neonates, infants, children, adults: $\leq$ 5%)

Respiratory: Apnea (3%), cough (neonates, infants, children, adults: $\leq$ 1%), respiratory depression ($\leq$ 7%), rhonchi (neonates, infants, children, adults: $\leq$ 3%), stridor (neonates, infants, children, adults: $\leq$ 1%)

<1%:

Cardiovascular: Cardiac arrhythmia, heart block, increased serum creatine kinase, ischemic heart disease, syncope, tachycardia

Dermatologic: Skin rash, urticaria

Endocrine & metabolic: Electrolyte disorder, hyperglycemia

Gastrointestinal: Abdominal distress, constipation, diarrhea, dysphagia, gastroesophageal reflux disease, hiccups, intestinal obstruction, xerostomia

Genitourinary: Dysuria, oliguria, urinary incontinence, urinary retention

Hematologic: Anemia, leukocytosis, lymphocytopenia, thrombocytopenia

Hepatic: Abnormal liver function

Hypersensitivity: Anaphylaxis

Local: Infusion-site reaction (including erythema at injection site, injection-site pruritus, rash at injection site)

Nervous system: Amnesia, anxiety, awareness under anesthesia without pain, confusion, decreased body temperature, delayed recovery from neuromuscular block, disorientation, dysphoria, hallucination, involuntary body movements, nightmares, paresthesia, prolonged emergence from anesthesia, rapid awakening from anesthesia, seizure, tremor, twitching

Neuromuscular & skeletal: Increased creatine phosphokinase in blood specimen, laryngospasm, musculoskeletal chest pain

Ophthalmic: Nystagmus disorder

Respiratory: Bronchitis, bronchospasm, dyspnea, hypoxia, nasal congestion, pharyngitis, pleural effusion, pulmonary edema, rales, rhinorrhea

Miscellaneous: Fever

Frequency not defined: Nervous system: Drug abuse, opioid dependence

Postmarketing:

Cardiovascular: Asystole

Nervous system: Allodynia (opioid-induced hyperalgesia) (FDA Safety Communication 2023)

Contraindications

Hypersensitivity (eg, anaphylaxis) to remifentanyl or any component of the formulation; intrathecal or epidural administration.

Canadian labeling: Additional contraindications (not in US labeling): Known or suspected mechanical GI obstruction (eg, bowel obstruction or strictures) or any diseases/conditions that affect bowel transit (eg, ileus of any type); suspected surgical abdomen (eg, acute appendicitis or pancreatitis); mild pain that can be managed with other pain medications; acute or severe bronchial asthma, chronic obstructive airway, status asthmaticus; acute respiratory depression; hypercapnia; cor pulmonale; acute alcoholism, delirium tremens, and convulsive disorders; severe CNS depression, increased cerebrospinal or intracranial pressure and head injury; concurrent use or use within 14 days of an MAO inhibitor

Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Warnings/Precautions

Concerns related to adverse effects:

- Esophageal dysfunction: Opioid-induced esophageal dysfunction has been reported; risk increases with higher doses and prolonged therapy.
- Hyperalgesia: Opioid-induced hyperalgesia (OIH) has occurred with short-term and prolonged use of opioid analgesics. Symptoms may include increased levels of pain upon opioid dosage increase, decreased levels of pain upon opioid dosage decrease, or pain from ordinarily nonpainful stimuli; symptoms may be suggestive of OIH if there is no evidence of underlying disease progression, opioid tolerance, opioid withdrawal, or addictive behavior. Consider decreasing the current opioid dose or opioid rotation in patients who experience OIH.
- Hypotension: May cause hypotension; use with caution in patients with hypovolemia, cardiovascular disease (including acute MI), or drugs which may exaggerate hypotensive effects (including

phenothiazines or general anesthetics). Monitor for symptoms of hypotension following initiation or dose titration.

- Intraoperative awareness: Intraoperative awareness has been reported when used with propofol infusion rates of ≤ 75 mcg/kg/minute in patients < 55 years.
- Respiratory depression: Fatal respiratory depression may occur. Carbon dioxide retention from opioid-induced respiratory depression can exacerbate the sedating effects of opioids. Have resuscitative and intubation equipment, oxygen, and an opioid overdose reversal agent (eg, naloxone, nalmefene) readily available.
- Serotonin syndrome: Potentially life-threatening serotonin syndrome (SS) has occurred with concomitant use of remifentanyl and serotonergic agents (eg, SSRIs, SNRIs, triptans, TCAs, 5-HT₃ receptor antagonists, mirtazapine, trazodone, tramadol) and agents that impair metabolism of serotonin (eg, MAO inhibitors). Monitor patients closely for signs of SS such as mental status changes (eg, agitation, hallucinations, delirium, coma); autonomic instability (eg, tachycardia, labile blood pressure, diaphoresis); neuromuscular changes (eg, tremor, rigidity, myoclonus); GI symptoms (eg, nausea, vomiting, diarrhea); and/or seizures. Discontinue remifentanyl if serotonin syndrome is suspected.

Disease-related concerns:

- Abdominal conditions: May obscure diagnosis or clinical course of patients with acute abdominal conditions.
- Biliary tract impairment: Use with caution in patients with biliary tract dysfunction, including acute pancreatitis; opioids may cause constriction of sphincter of Oddi.
- Bradycardia: Use with caution when administering to patients with bradycardia.
- Delirium tremens: Use with caution in patients with delirium tremens.
- Head trauma: Use with extreme caution in patients with head injury, intracranial lesions, or elevated intracranial pressure (ICP); exaggerated elevation of ICP may occur.
- Obesity: Use with caution in patients who are morbidly obese.
- Prostatic hyperplasia/urinary stricture: Use with caution in patients with prostatic hyperplasia and/or urinary stricture.
- Psychosis: Use with caution in patients with toxic psychosis.
- Respiratory disease: Use with caution and monitor for respiratory depression in patients with significant chronic obstructive pulmonary disease or cor pulmonale, and those with a substantially decreased respiratory reserve, hypoxia, hypercapnia, or preexisting respiratory depression, particularly when initiating and titrating therapy; critical respiratory depression may occur, even at therapeutic dosages.
- Seizures: Use with caution in patients with a history of seizure disorders; may increase risk or exacerbate preexisting seizure disorders.
- Thyroid dysfunction: Use with caution in patients with thyroid dysfunction.

Special populations:

- Cachectic or debilitated patients: Use with caution in cachectic or debilitated patients; there is a greater potential for critical respiratory depression, even at therapeutic dosages.
- Older adult: Use with caution in older adults; may be more sensitive to adverse effects. Decrease initial dose.

Other warnings/precautions:

- Abuse/misuse/diversion: Use with caution in patients with a history of substance use disorder; potential for drug dependency exists. Other factors associated with increased risk for misuse include concomitant depression or other mental health conditions, higher opioid dosages, or taking other CNS depressants.

- **Appropriate use:** Inadequate clearing of IV tubing following administration has been associated with muscle rigidity, respiratory depression, and apnea when another fluid is administered through the same line. Do not administer into the same IV tubing with blood due to potential inactivation by nonspecific esterases in blood products.
- **Discontinuation of therapy:** Interruption of an infusion will result in offset of effect within 5 to 10 minutes; the discontinuation of an infusion should be preceded by the establishment of adequate postoperative analgesia. Do not rapidly reduce or abruptly discontinue in patients with substance use disorder; withdrawal syndrome may occur.
- **General anesthesia use:** Not recommended as the sole agent for induction of anesthesia, because the loss of consciousness cannot be assured.
- **Rapid infusion:** Rapid IV infusion (single dose >1 mcg/kg over 30 to 60 seconds and infusion rates >0.1 mcg/kg/minute) should only be used during maintenance of general anesthesia; rapid infusion may result in skeletal muscle and chest wall rigidity. Chest wall rigidity may resolve by decreasing the infusion rate, discontinuing the infusion, or by administering a neuromuscular blocking agent.
- **Trained individuals:** Remifentanyl should only be administered by health care providers specifically trained in the use of anesthetic agents. Should not be used in diagnostic or therapeutic procedures outside the monitored anesthesia setting; resuscitative and intubation equipment should be readily available.

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution Reconstituted, Intravenous:

Ultiva: 1 mg (1 ea)

Generic: 2 mg (1 ea)

Solution Reconstituted, Intravenous [preservative free]:

Ultiva: 1 mg (1 ea); 2 mg (1 ea); 5 mg (1 ea)

Generic: 1 mg (1 ea); 2 mg (1 ea); 5 mg (1 ea)

Generic Equivalent Available: US

Yes

Pricing: US

Solution (reconstituted) (Remifentanyl HCl Intravenous)

1 mg (per each): \$50.40 - \$87.97

2 mg (per each): \$147.00 - \$154.34

5 mg (per each): \$312.00 - \$407.95

Solution (reconstituted) (Ultiva Intravenous)

1 mg (per each): \$80.83

2 mg (per each): \$161.66

5 mg (per each): \$343.96

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind

or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution Reconstituted, Intravenous:

Generic: 1 mg (1 ea); 2 mg (1 ea)

Controlled Substance

C-II

Administration: Adult

IV: For IV use only. Must be diluted prior to administration. An infusion device should be used to administer continuous infusions. During the maintenance of general anesthesia, IV boluses may be administered over 30 to 60 seconds. Injections should be given into IV tubing close to the venous cannula; tubing should be cleared after treatment to prevent residual effects when other fluids are administered through the same IV line.

Preparation for Administration: Adult

Add 1 mL of D5LR, D5NS, D5W, $\frac{1}{2}$ NS, NS, or SWFI per 1 mg of remifentanyl (resulting solution: 1 mg/mL). Shake well. Further dilute to a final concentration of 20, 25, 50, or 250 mcg/mL.

Administration: Pediatric

IV: An infusion pump should be used to administer continuous infusions. During the maintenance of general anesthesia, IV boluses ≤ 1 mcg/kg may be administered over 30 to 60 seconds; for doses > 1 mcg/kg, administer over > 60 seconds (to reduce the potential to develop skeletal muscle and chest wall rigidity). Injections should be given into IV tubing close to the venous cannula; tubing should be cleared after treatment to prevent residual effects when other fluids are administered through the same IV line.

Preparation for Administration: Pediatric

IV: Prepare solution by adding 1 mL of diluent per 1 mg of remifentanyl. Shake well. Further dilute to a final concentration of 20, 25, 50, or 250 mcg/mL.

Storage/Stability

Store intact vials at 2°C to 25°C (36°F to 77°F). Stable for 24 hours at room temperature after reconstitution and further dilution in SWFI, D5LR, D5NS, D5W, $\frac{1}{2}$ NS, or NS to concentrations of 20 to 250 mcg/mL (4 hours if diluted with LR).

Use: Labeled Indications

General anesthesia: Analgesic for use during the induction and maintenance of general anesthesia for inpatient and outpatient procedures; continued analgesia into the immediate postoperative period in adults under the direct supervision of an anesthesia practitioner in a postoperative anesthesia care unit or intensive care setting; analgesic component of monitored anesthesia in adults.

Use: Off-Label: Adult

Mechanically ventilated patients in the ICU, pain and sedation

Medication Safety Issues

Sound-alike/look-alike issues:

Remifentanyl may be confused with Alfentanil

High alert medication:

The Institute for Safe Medication Practices (ISMP) includes this medication among its list of drug classes (opioids, all formulations and routes of administration) which have a heightened risk of causing significant patient harm when used in error (ISMP List of High-Alert Medications in Acute Care Settings ©ISMP 2024; ISMP List of High-Alert Medications in Community/Ambulatory Care Settings ©ISMP 2021; High-Alert Medications in Long-Term Care (LTC) Settings ©ISMP 2021).

Metabolism/Transport Effects

None known.

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within “CYP3A4 Inducers [Strong]” are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the “Launch drug interactions program” link above.

Agents with Clinically Relevant Anticholinergic Effects: May increase adverse/toxic effects of Opioid Agonists. Specifically, the risk for constipation and urinary retention may be increased with this combination. *Risk C: Monitor*

Alcohol (Ethyl): May increase CNS depressant effects of Opioid Agonists. *Risk X: Avoid*

Alfuzosin: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Alizapride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Allopurinol: CNS Depressants may increase sedative effects of Allopurinol. *Risk C: Monitor*

Alvimopan: Opioid Agonists may increase adverse/toxic effects of Alvimopan. This is most notable for patients receiving long-term (i.e., more than 7 days) opiates prior to alvimopan initiation. Management: Alvimopan is contraindicated in patients receiving therapeutic doses of opioids for more than 7 consecutive days immediately prior to alvimopan initiation. *Risk D: Consider Therapy Modification*

Amifostine: Blood Pressure Lowering Agents may increase hypotensive effects of Amifostine. Management: When used at chemotherapy doses, hold blood pressure lowering medications for 24 hours before amifostine administration. If blood pressure lowering therapy cannot be held, do not administer amifostine. Use caution with radiotherapy doses of amifostine. *Risk D: Consider Therapy Modification*

Amisulpride (Oral): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Amisulpride (Oral): May increase hypotensive effects of Hypotension-Associated Agents. *Risk C: Monitor*

Amphetamines: May increase analgesic effects of Opioid Agonists. *Risk C: Monitor*

Antipsychotic Agents (Second Generation [Atypical]): Blood Pressure Lowering Agents may increase hypotensive effects of Antipsychotic Agents (Second Generation [Atypical]). *Risk C: Monitor*

Arginine: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Azelastine (Nasal): May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Barbiturates: CNS Depressants may increase CNS depressant effects of Barbiturates. *Risk C: Monitor*

Benperidol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Benperidol: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Blood Pressure Lowering Agents: May increase hypotensive effects of Hypotension-Associated Agents. *Risk C: Monitor*

Bradycardia-Causing Agents: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Brimonidine (Topical): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Brimonidine (Topical): May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Bromopride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Bromperidol: May decrease hypotensive effects of Blood Pressure Lowering Agents. Blood Pressure Lowering Agents may increase hypotensive effects of Bromperidol. *Risk X: Avoid*

Buprenorphine: CNS Depressants may increase CNS depressant effects of Buprenorphine. Management: Avoid coadministration of buprenorphine and other CNS depressants if possible. If combined, consider adjustments to induction, increased monitoring, and co-prescription of an opioid overdose reversal agent. *Risk D: Consider Therapy Modification*

Buprenorphine: May decrease therapeutic effects of Opioid Agonists. Management: Seek alternatives to buprenorphine in patients receiving pure opioid agonists. If combined in certain pain management situations (eg, surgery), monitor for symptoms of therapeutic failure/high dose requirements or opioid withdrawal symptoms. *Risk D: Consider Therapy Modification*

BusPIRone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Carbinoxamine: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Ceritinib: Bradycardia-Causing Agents may increase bradycardic effects of Ceritinib. Management: If this combination cannot be avoided, monitor patients for evidence of symptomatic bradycardia, and closely monitor blood pressure and heart rate during therapy. *Risk D: Consider Therapy Modification*

Chlorphenesin Carbamate: May increase adverse/toxic effects of CNS Depressants. *Risk C: Monitor*

ChlorproMAZINE: May increase CNS depressant effects of Opioid Agonists. Management: Avoid concomitant use of opioid agonists and chlorpromazine when possible. These agents should only be combined if alternative treatment options are inadequate. If combined, limit the dosage and duration of each drug. *Risk D: Consider Therapy Modification*

Cimetidine: May increase analgesic effects of Opioid Agonists. Cimetidine may increase respiratory depressant effects of Opioid Agonists. *Risk C: Monitor*

CNS Depressants: May increase CNS depressant effects of Opioid Agonists. Management: Avoid concomitant use of opioid agonists and benzodiazepines or other CNS depressants when possible. These agents should only be combined if alternative treatment options are inadequate. If combined, limit the dosage and duration of each drug. *Risk D: Consider Therapy Modification*

Desmopressin: Opioid Agonists may increase hyponatremic effects of Desmopressin. *Risk C: Monitor*

Diazoxide (Immediate Release): May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Difelikefalin: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Dihydralazine: CNS Depressants may increase hypotensive effects of Dihydralazine. *Risk C: Monitor*

Dimethindene (Topical): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Diuretics: Opioid Agonists may increase adverse/toxic effects of Diuretics. Opioid Agonists may decrease therapeutic effects of Diuretics. *Risk C: Monitor*

DULoxetine: Blood Pressure Lowering Agents may increase hypotensive effects of DULoxetine. *Risk C: Monitor*

Eluxadolone: Opioid Agonists may increase constipating effects of Eluxadolone. *Risk X: Avoid*

Emedastine (Systemic): May increase CNS depressant effects of CNS Depressants. Management: Consider avoiding this combination if possible. If required, monitor for excessive sedation or CNS

depression, limit the dose and duration of combination therapy, and consider CNS depressant dose reductions. *Risk C: Monitor*

Entacapone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Etrasimod: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Fexinidazole: Bradycardia-Causing Agents may increase arrhythmogenic effects of Fexinidazole. *Risk X: Avoid*

Fingolimod: Bradycardia-Causing Agents may increase bradycardic effects of Fingolimod. Management: Consult with the prescriber of any bradycardia-causing agent to see if the agent could be switched to an agent that does not cause bradycardia prior to initiating fingolimod. If combined, perform continuous ECG monitoring after the first fingolimod dose. *Risk D: Consider Therapy Modification*

Flunarizine: CNS Depressants may increase CNS depressant effects of Flunarizine. *Risk X: Avoid*

Gastrointestinal Agents (Prokinetic): Opioid Agonists may decrease therapeutic effects of Gastrointestinal Agents (Prokinetic). *Risk C: Monitor*

Herbal Products with Blood Pressure Lowering Effects: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Hypotension-Associated Agents: Blood Pressure Lowering Agents may increase hypotensive effects of Hypotension-Associated Agents. *Risk C: Monitor*

Immune Checkpoint Inhibitors (Anti-PD-1, -PD-L1, and -CTLA4 Therapies): May decrease therapeutic effects of Opioid Agonists. Opioid Agonists may decrease therapeutic effects of Immune Checkpoint Inhibitors (Anti-PD-1, -PD-L1, and -CTLA4 Therapies). *Risk C: Monitor*

Inhalational Anesthetics: CNS Depressants may increase CNS depressant effects of Inhalational Anesthetics. *Risk C: Monitor*

Isocarboxazid: CNS Depressants may increase adverse/toxic effects of Isocarboxazid. *Risk X: Avoid*

Ivabradine: Bradycardia-Causing Agents may increase bradycardic effects of Ivabradine. *Risk C: Monitor*

Ixabepilone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Kava Kava: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Ketotifen (Systemic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Kratom: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Landiolol: Bradycardia-Causing Agents may increase bradycardic effects of Landiolol. *Risk X: Avoid*

Levocetirizine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Levodopa-Foslevodopa: Blood Pressure Lowering Agents may increase hypotensive effects of Levodopa-Foslevodopa. *Risk C: Monitor*

Lisuride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Lofexidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Magnesium Sulfate: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Mequitazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Metergoline: May decrease antihypertensive effects of Blood Pressure Lowering Agents. Blood Pressure Lowering Agents may increase orthostatic hypotensive effects of Metergoline. *Risk C: Monitor*

Metergoline: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Methotrimeprazine: May increase CNS depressant effects of Opioid Agonists. Management: Avoid concomitant use of opioid agonists and methotrimeprazine when possible. These agents should only

be combined if alternative treatment options are inadequate. If combined, limit the dosage and duration of each drug. *Risk D: Consider Therapy Modification*

Metoclopramide: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

MetyroSINE: CNS Depressants may increase sedative effects of MetyroSINE. *Risk C: Monitor*

Midodrine: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Minocycline (Systemic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Molsidomine: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Monoamine Oxidase Inhibitors: Remifentanyl may increase serotonergic effects of Monoamine Oxidase Inhibitors. This could result in serotonin syndrome. Monoamine Oxidase Inhibitors may increase adverse/toxic effects of Remifentanyl. Specifically, the risk for opioid toxicity (eg, respiratory depression) may be increased. Management: The use of remifentanyl is not recommended for patients taking monoamine oxidase inhibitors (MAOIs) or within 14 days of MAOI discontinuation. If coadministration is required, use test doses and titrate small doses of remifentanyl frequently. *Risk D: Consider Therapy Modification*

Moxonidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Nabilone: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Naftopidil: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Nalfurafine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Nalfurafine: Opioid Agonists may increase adverse/toxic effects of Nalfurafine. Opioid Agonists may decrease therapeutic effects of Nalfurafine. *Risk C: Monitor*

Nalmefene: May decrease therapeutic effects of Opioid Agonists. Management: Avoid the concomitant use of oral nalmefene and opioid agonists. Discontinue oral nalmefene 1 week prior to any anticipated use of opioid agonists. If combined, larger doses of opioid agonists will likely be required. *Risk D: Consider Therapy Modification*

Naltrexone: May increase or decrease therapeutic effects of Opioid Agonists. Management: Seek therapeutic alternatives to opioids. See full drug interaction monograph for detailed recommendations. *Risk X: Avoid*

Nicergoline: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Nicorandil: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Nitroprusside: Blood Pressure Lowering Agents may increase hypotensive effects of Nitroprusside. *Risk C: Monitor*

Noscapine: CNS Depressants may increase adverse/toxic effects of Noscapine. *Risk X: Avoid*

Obinutuzumab: May increase hypotensive effects of Blood Pressure Lowering Agents. Management: Consider temporarily withholding blood pressure lowering medications beginning 12 hours prior to obinutuzumab infusion and continuing until 1 hour after the end of the infusion. *Risk D: Consider Therapy Modification*

Olopatadine (Nasal): May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Opicapone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Opioid Agonists: CNS Depressants may increase CNS depressant effects of Opioid Agonists. Management: Avoid concomitant use of opioid agonists and benzodiazepines or other CNS depressants when possible. These agents should only be combined if alternative treatment options are inadequate. If combined, limit the dosage and duration of each drug. *Risk D: Consider Therapy Modification*

Opioids (Mixed Agonist / Antagonist): May decrease analgesic effects of Opioid Agonists. Management: Seek alternatives to mixed agonist/antagonist opioids in patients receiving pure opioid ag-

onists, and monitor for symptoms of therapeutic failure/high dose requirements (or withdrawal in opioid-dependent patients) if patients receive these combinations. *Risk X: Avoid*

Opi Pramol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Oxomemazine: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Ozanimod: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Paraldehyde: CNS Depressants may increase CNS depressant effects of Paraldehyde. *Risk X: Avoid*

Pegvisomant: Opioid Agonists may decrease therapeutic effects of Pegvisomant. *Risk C: Monitor*

Pentoxifylline: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Periciazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Phosphodiesterase 5 Inhibitors: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Pipamperone: May increase adverse/toxic effects of CNS Depressants. *Risk C: Monitor*

Piribedil: CNS Depressants may increase CNS depressant effects of Piribedil. *Risk C: Monitor*

Ponesimod: Bradycardia-Causing Agents may increase bradycardic effects of Ponesimod. Management: Avoid coadministration of ponesimod with drugs that may cause bradycardia when possible. If combined, monitor heart rate closely and consider obtaining a cardiology consult. Do not initiate ponesimod in patients on beta-blockers if HR is less than 55 bpm. *Risk D: Consider Therapy Modification*

Pramipexole: CNS Depressants may increase sedative effects of Pramipexole. *Risk C: Monitor*

Procarbazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Promazine: CNS Depressants may increase CNS depressant effects of Promazine. *Risk C: Monitor*

Prostacyclin Analogues: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Ramosetron: Opioid Agonists may increase constipating effects of Ramosetron. *Risk C: Monitor*

Rilmenidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

ROPINIrole: CNS Depressants may increase sedative effects of ROPINIrole. *Risk C: Monitor*

Rotigotine: CNS Depressants may increase sedative effects of Rotigotine. *Risk C: Monitor*

Samidorphan: May decrease therapeutic effects of Opioid Agonists. Specifically, the risk for opioid withdrawal may be increased. *Risk X: Avoid*

Serotonergic Agents (High Risk): Opioid Agonists may increase serotonergic effects of Serotonergic Agents (High Risk). This could result in serotonin syndrome. Management: Monitor for signs and symptoms of serotonin syndrome/serotonin toxicity (eg, hyperreflexia, clonus, hyperthermia, diaphoresis, tremor, autonomic instability, mental status changes) when these agents are combined. *Risk C: Monitor*

Silodosin: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Sincalide: Drugs that Affect Gallbladder Function may decrease therapeutic effects of Sincalide. Management: Consider discontinuing drugs that may affect gallbladder motility prior to the use of sincalide to stimulate gallbladder contraction. *Risk D: Consider Therapy Modification*

Siponimod: Bradycardia-Causing Agents may increase bradycardic effects of Siponimod. Management: Avoid coadministration of siponimod with drugs that may cause bradycardia. If combined, consider obtaining a cardiology consult regarding patient monitoring. *Risk D: Consider Therapy Modification*

Somatostatin Analogs: May increase or decrease analgesic effects of Opioid Agonists. *Risk C: Monitor*

Succinylcholine: May increase bradycardic effects of Opioid Agonists. *Risk C: Monitor*

Thalidomide: CNS Depressants may increase CNS depressant effects of Thalidomide. *Risk X: Avoid*

Tilidine: May increase therapeutic effects of Opioid Agonists. *Risk X: Avoid*

Tradipitant: CNS Depressants may increase CNS depressant effects of Tradipitant. *Risk C: Monitor*

Valerian: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Zolpidem: May increase CNS depressant effects of Opioid Agonists. Management: Avoid concomitant use of opioid agonists and zolpidem when possible. These agents should only be combined if alternative treatment options are inadequate. If combined, limit the dosage and duration of each drug. *Risk D: Consider Therapy Modification*

Pregnancy Considerations

Remifentanil crosses the placenta. In trials, maternal remifentanil concentrations were generally observed to be twice fetal concentration; however, similar fetal and maternal concentrations have also been reported. Umbilical arteriovenous concentration analysis suggests neonatal metabolism.

Prolonged use of opioids during pregnancy can cause neonatal withdrawal syndrome, which may be life-threatening if not recognized and treated according to protocols developed by neonatology experts. Opioids used as part of obstetric analgesia/anesthesia during labor and delivery may temporarily affect the fetal heart rate (ACOG 209 2019). Opioids may cause respiratory depression and psychophysiologic effects in the neonate; newborns of mothers receiving remifentanil during labor should be closely monitored (Devroe 2015; Noskova 2015); have an opioid overdose reversal agent (eg, naloxone, nalmefene) available for reversal of opioid-induced respiratory depression in the neonate.

Pharmacokinetic properties of remifentanil are not significantly altered by pregnancy; dosing adjustment is not required for labor analgesia or immediate postpartum use (ACOG 209 2019; Devroe 2015; Smith 2017; Weibel 2017).

The ACOG recommends that pregnant women should not be denied medically necessary surgery, regardless of trimester. If the procedure is elective, it should be delayed until after delivery (ACOG 775 2019).

Breastfeeding Considerations

It is not known if remifentanil is present in breast milk.

According to the manufacturer, the decision to continue or discontinue breastfeeding during therapy should consider the risk of infant exposure, the benefits of breastfeeding to the infant, and benefits of treatment to the mother; monitor infants for excess sedation and respiratory depression. Remifentanil has a limited duration of action; use may be appropriate for breastfeeding women undergoing short procedures (ABM [Reece-Stremtan 2017]).

The Academy of Breast Feeding Medicine recommends postponing elective surgery until milk supply and breastfeeding are established. Milk should be expressed ahead of surgery when possible. In general, when the child is healthy and full term, breastfeeding may resume, or milk may be expressed once the mother is awake and in recovery. For children who are at risk for apnea, hypotension, or hypotonia, milk may be saved for later use when the child is at lower risk (ABM [Reece-Stremtan 2017]).

Monitoring Parameters

Respiratory and cardiovascular status, blood pressure, heart rate; signs or symptoms of opioid-induced esophageal dysfunction (eg, chest pain [noncardiac], dysphagia, regurgitation).

Critically ill: The Numeric Rating Scale should be used in patients who are able to self-report pain. In patients who are unable to self-report pain, the Behavioral Pain Scale and the Critical-Care Pain Observational Tool can be used in intubated or nonintubated patients (SCCM [Devlin 2018]).

Mechanism of Action

Binds with stereospecific mu-opioid receptors at many sites within the CNS, increases pain threshold, alters pain reception, inhibits ascending pain pathways

Pharmacokinetics (Adult Data Unless Noted)

Note: In pediatric patients (neonates through adolescents), pharmacokinetic data showed variable age-related changes with distribution (ie, highest V_d associated with lowest age) and clearance (ie, fastest clearance in the youngest patients); however, no age-related changes with half-life (Ross 2001).

Onset of action: IV: 1 to 3 minutes; Peak effect: 3 to 5 minutes

Duration: 3 to 10 minutes (Scott 2005)

Distribution:

Pediatric patients (Ross 2001): V_{dss} :

Neonates ≤ 2 months: 453 ± 145 mL/kg

Infants and Children > 2 months to < 2 years: 308 ± 89 mL/kg

Children 2 to 6 years: 240 mL/kg ± 131 mL/kg

Children 7 to 12 years: 249 mL/kg ± 91 mL/kg

Adolescents 13 to < 16 years: 223 ± 31 mL/kg

Adolescents 16 to 18 years: 243 ± 109 mL/kg

Adults: V_d : Initial: 100 mL/kg; V_{dss} : 350 mL/kg

Protein binding: $\sim 70\%$ (primarily alpha-1 acid glycoprotein)

Metabolism: Rapid via blood and tissue esterases; not metabolized by plasma cholinesterase (pseudocholinesterase) and is not appreciably metabolized by the liver

Half-life elimination (dose dependent):

Pediatric patients (Ross 2001): Effective:

Neonates ≤ 2 months: 5.4 minutes (range: 3 to 8 minutes)

Infants and Children > 2 months to < 2 years: 3.4 minutes (range: 2 to 6 minutes)

Children 2 to 6 years: 3.6 minutes (range: 1 to 6 minutes)

Children 7 to 12 years: 5.3 minutes (range: 3 to 7 minutes)

Adolescents: 13 to < 16 years: 3.7 minutes (range: 2 to 5 minutes)

Adolescents 16 to 18 years: 5.7 minutes (range: 5 to 6 minutes)

Adults: Terminal: 10 to 20 minutes; Effective: 3 to 10 minutes

Time to peak, serum: Intranasal: Children ≤ 7 years: ~ 3.5 minutes (Verghese 2008)

Excretion: Urine

Clearance:

Pediatric patients (Ross 2001):

Neonates ≤ 2 months of age: 90.5 ± 36.8 mL/minute/kg

Infants and Children > 2 months to < 2 years: 92.1 ± 25.8 mL/minute/kg

Children 2 to 6 years: 76.0 ± 22.4 mL/minute/kg

Children 7 to 12 years: 59.7 ± 22.5 mL/minute/kg

Adolescents 13 to < 16 years: 57.2 ± 21.2 mL/minute/kg

Adolescents 16 to 18 years: 46.5 ± 2.1 mL/minute/kg

Adults: ~40 mL/minute/kg

Pharmacokinetics: Additional Considerations (Adult Data Unless Noted)

Older adult: Clearance is reduced ~25%.

Patient Counseling Points

(For additional information see "Remifentanyl: Patient drug information")

What is this drug used for?

- It is used to ease pain.
- It is used during surgery.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

- Upset stomach or throwing up
- Shivering
- Dizziness or headache
- Itching
- Sweating a lot
- Constipation

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

- High or low blood pressure like very bad headache or dizziness, passing out, or change in eyesight
- A heartbeat that does not feel normal
- Slow heartbeat
- Fever
- Muscle stiffness
- Seizures
- Breathing problems like slow, shallow, or trouble breathing
- Serotonin syndrome like agitation; change in balance; confusion; hallucinations; fever; fast or abnormal heartbeat; flushing; muscle twitching or stiffness; seizures; shivering or shaking; sweating a lot; severe diarrhea, upset stomach, or throwing up; or severe headache
- Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

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include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AE) United Arab Emirates: Ultiva;
- (AR) Argentina: Fada remifentanilo | Redormin | Remicit | Remifas | Remifentanilo gray | Remifentanilo kabi | Remifentanilo kilab | Remifentanilo northia | Remifentanilo pharmavial | Remifentanilo richet | Remiflo | Restinil | Sagal | Ultiva;
- (AT) Austria: Remifentanil | Remifentanil chiesi | Remifentanil kabi | Remifentanil sandoz | Remifentanil Teva | Ultiva;
- (AU) Australia: Apo remifentanil | Dbl remifentanil | Remifentanil aft | Remifentanil alphapharm | Remifentanil sandoz | Remifentanil viatris | Ultiva;
- (BE) Belgium: Remifentanil | Remifentanil actavis | Remifentanil Fresenius kabi | Remifentanil hospira | Remifentanil mylan | Remifentanil sandoz | Remifentanil Teva | Ultiva;
- (BG) Bulgaria: Remifentanil hameln | Remifentanil Tchaikapharma;
- (BR) Brazil: Cloridrato de remifentanila | Relyon | Remifas | Remistes | Ultiva;
- (CH) Switzerland: Remifentanil fresenius | Remifentanil sandoz | Remifentanil Teva;
- (CL) Chile: Fenburol | Remifas | Remifentanilo | Ultiva;
- (CN) China: Rui jie;
- (CO) Colombia: Rafentilo | Remifentafile | Remifentanilo | Remifentanilo b. braun | Serilone | Suremifen | Tenotalis | Ultiva;
- (CZ) Czech Republic: Remifentanil braun | Ultiva;
- (DE) Germany: Remifentanil | Remifentanil actavis | Remifentanil B.Bra | Remifentanil hameln | Remifentanil hexal | Remifentanil kabi | Remifentanil Teva | Remimed | Ultiva;
- (DO) Dominican Republic: Remifas;
- (EC) Ecuador: Clorhidrato de remifentanilo | Rafentilo | Remifas | Remifentanil | Remifentanilo | Restinil | Tenotalis | Ultiva;
- (EE) Estonia: Remifentanil kabi | Ultiva;
- (ES) Spain: Remifentanilo combino pharm | Remifentanilo kabi | Remifentanilo kern pharma | Remifentanilo Normon | Remifentanilo sala | Ultiva;
- (FI) Finland: Remifentanil actavis | Remifentanil b. braun | Remifentanil hameln | Remifentanil hospira | Remifentanil orion | Ultiva;
- (FR) France: Remifentanil actavis | Remifentanil b braun | Remifentanil hospira | Remifentanil mylan | Remifentanil Teva | Ultiva;
- (GB) United Kingdom: Remifentanil | Ultiva;
- (GR) Greece: Dormiden | Remifentanil/teva | Ultiva | Ultizen;
- (HK) Hong Kong: Ultiva;
- (HU) Hungary: Ultiva;
- (IE) Ireland: Remifentanil | Remifentanil actavis | Ultiva;
- (IT) Italy: Remifentanil accord | Remifentanil actavis | Remifentanil b. braun | Remifentanil hospira | Remifentanil kabi | Remifentanil mylan | Remifentanil orion | Remifentanil Teva | Ultiva;
- (JO) Jordan: Rimfanile | Ultiva;
- (JP) Japan: Remifentanil daiichi sankyo | Ultiva;
- (KE) Kenya: Ultiva;
- (KR) Korea, Republic of: Domiden | Kabi remifentanil | Remiva | Tivare | Ultian | Ultifen | Ultiva;
- (KW) Kuwait: Remifentanil b. braun;
- (LB) Lebanon: Ultiva;

- (LT) Lithuania: Remifentanil kabi | Ultiva;
- (LU) Luxembourg: Ultiva;
- (LV) Latvia: Remifentanil kabi | Ultiva;
- (MX) Mexico: Proximicil | Selatron | Ultiva;
- (MY) Malaysia: Ultiva;
- (NL) Netherlands: Remifentanil actavis | Remifentanil Fresenius kabi | Remifentanil hospira | Remifentanil mylan | Remifentanil Teva | Ultiva;
- (NO) Norway: Remifentanil | Remifentanil actavis | Remifentanil braun | Remifentanil hameln | Remifentanil orion | Ultiva;
- (NZ) New Zealand: Aft Remifentanil | Ultiva;
- (PE) Peru: Remifentanilo | Ultiva | Utivex;
- (PL) Poland: Remifentanil actavis | Remifentanil b.braun | Remifentanil chiesi | Remifentanil kabi | Ultiva;
- (PR) Puerto Rico: Ultiva;
- (PT) Portugal: Remifentanil | Remifentanilo | Remifentanilo actavis | Remifentanilo b. braun | Remifentanilo hospira | Remifentanilo kabi | Remifentanilo Sandoz | Remifentanilo teva | Ultiva;
- (PY) Paraguay: Remifentanilo kabi | Remifentanilo quimfa | Restinil | Sagal;
- (QA) Qatar: Dormiden | Lentanil | Rentanil | Ultiva;
- (RO) Romania: Remifentanil kabi | Ultiva;
- (SA) Saudi Arabia: Remifentanil medis;
- (SE) Sweden: Remifentanil actavis | Remifentanil b.braun | Remifentanil Fresenius kabi | Remifentanil hameln | Remifentanil orion | Remifentanil reig jofre | Remifentanil sandoz | Remifentanil Teva | Ultiva;
- (SG) Singapore: Ultiva;
- (SI) Slovenia: Remifentanil chiesi | Remifentanil hameln | Remifentanil orion | Remifentanil sandoz | Remifentanil Teva | Remifentanilo kern pharma | Ultiva;
- (SK) Slovakia: Remifentanil braun | Remifentil Chiesi | Ultiva;
- (SR) Suriname: Ultiva;
- (SV) El Salvador: Remifentanil | Remiflo;
- (TN) Tunisia: Remifentanil medis | Remifentanil mylan | Ultiva;
- (TR) Turkey: Opiva | Reksiva | Remanest | Rentanil | Ultan | Ultiva;
- (TW) Taiwan: Remifentanil alvogen;
- (UA) Ukraine: Remifentanil vista;
- (UG) Uganda: Ultiva;
- (UY) Uruguay: Fada remifentanilo | Remifas | Remifentanilo kabi | Rentil | Restinil | Ultiva;
- (VE) Venezuela, Bolivarian Republic of: Remifentanilo | Ultiva;
- (ZA) South Africa: Altirem IV | Mylan remifentanil | Remifentanil fresenius | Ultiva

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